

E-Commerce SQL Portfolio Project

Full List of 150 SQL Practice Questions — Customer Analytics Database

SECTION 1: Basic Queries (1–55)

Single-table SELECT statements, filtering, sorting, basic aggregate functions, and GROUP BY operations.

1. Categories Table
2. Customers Table
3. Employees Table
4. Orderitems Table
5. Orders Table
6. Products Table
7. Reviews Table
8. Suppliers Table
9. Get the distinct countries customers live in.
10. Get the distinct cities suppliers operate from.
11. Find products priced above \$500
12. Find customers from the USA.
13. Find orders with status 'Delivered'.
14. Find employees in the IT department.
15. Find products with stock quantity below 100.
16. Top 10 most expensive products.
17. Customers who signed up in 2024.
18. Orders placed after Jan 1, 2025.
19. Reviews with a 5-star rating.
20. Employees hired before 2020.
21. Products that are discontinued.
22. Count total number of customers.
23. Count total number of products.
24. Count total number of orders.
25. Count total number of employees.
26. Average unit price of all products.
27. Maximum salary among employees.
28. Minimum salary among employees.
29. Total shipping cost collected across all orders.

30. Count of orders grouped by status.
31. Count of customers grouped by country.
32. Count of products grouped by category_id.
33. Average product price grouped by category.
34. Customers whose first name starts with 'A'.
35. Products with 'Pro' in the name.
36. Customers not from the USA.
37. Orders placed between January 2024 and December 2024.
38. Products priced between 100 and 500.
39. Employees who have no manager (top-level).
40. Orders that haven't shipped yet.
41. Get the first 5 customers by ID.
42. Get the 5 most recently hired employees.
43. List all distinct order statuses.
44. List all distinct departments.
45. Suppliers based in Japan or Germany.
46. Products that are still active (not discontinued).
47. Get email addresses of all customers.
48. Count the number of distinct categories.
49. Count the number of distinct suppliers actually used by products.
50. Total quantity ordered across all order items.
51. Average discount applied across all order items.
52. Employees ordered by salary, highest first.
53. Products ordered by stock quantity, lowest first.
54. Reviews mentioning the word 'recommend'.
55. Customers whose phone number contains a specific area code pattern.

SECTION 2: Intermediate Queries (56–105)

Multi-table joins, HAVING, subqueries, CASE statements, and date/string manipulation.

56. List orders along with the customer's name.
57. List order items along with the product name.
58. List products along with their category name.
59. List products along with their supplier name.
60. List each employee with their manager's name (self join).
61. List orders along with the employee who processed them.

62. List reviews with product name and customer name (two joins).
63. Total revenue per order (accounting for discount).
64. Total amount spent per customer.
65. Customers with more than 5 orders.
66. Number of orders handled per employee.
67. Products with an average rating above 4.
68. Total quantity sold per product.
69. Categories with more than 50 products.
70. Customers with more than 10 orders.
71. Products that have never been ordered.
72. Customers who have never placed an order.
73. Products with no reviews.
74. Employees who have processed more than 100 orders.
75. Orders placed by the single highest-spending customer.
76. Products priced above the overall average product price.
77. Customers who spent more than the average customer's total spend.
78. Find the second highest priced product.
79. Products supplied by suppliers based in the USA.
80. Categorize orders by shipping cost (Low/Medium/High) using CASE.
81. Categorize products into price bands using CASE.
82. Number of orders placed per year.
83. New customer sign-ups per month.
84. Orders placed in the last 30 days (relative to the most recent order).
85. Employee tenure in years.
86. Customers whose email domain is 'mail.com'.
87. Full name of each customer using CONCAT.
88. Product names in uppercase and lowercase.
89. Length of each product name.
90. Orders with total order value computed via a correlated subquery.
91. Top 3 categories by total revenue.
92. Top 5 products by total revenue.
93. Monthly revenue trend.
94. Total revenue by customer country.
95. Total revenue generated per employee (sales performance).

- 96. Average order value (AOV) per customer.
- 97. Customers who have at least one cancelled order.
- 98. Discontinued products that still have reviews.
- 99. Orders with shipping cost above the overall average shipping cost.
- 100. Managers and how many direct reports each one has.
- 101. Count of reviews for each rating value (1–5).
- 102. Percentage of orders in each status.
- 103. Suppliers who supply products across more than 3 different categories.
- 104. Categories whose average product price is above the overall average.
- 105. Orders that contain more than 3 order items (large basket orders).

SECTION 3: Advanced Queries (106–150)

Window functions, CTEs, recursive CTEs, market basket analysis, cohort & RFM analytics, and views.

- 106. Rank products by price within each category.
- 107. Assign a sequence number to each customer's orders (order 1, 2, 3...).
- 108. Running total of daily revenue.
- 109. Top 3 highest-revenue products within each category.
- 110. Each customer's first and most recent order date.
- 111. Month-over-month revenue growth percentage.
- 112. Days between each customer's consecutive orders.
- 113. Each customer's next order date (LEAD).
- 114. Cumulative count of new customers by signup month.
- 115. Percentile rank of each product's price.
- 116. Divide customers into 4 spending quartiles (NTILE).
- 117. 7-day moving average of daily revenue.
- 118. Pivot order counts by status into columns (conditional aggregation).
- 119. Pivot monthly revenue into columns per year (Jan–Dec across columns).
- 120. Customers who haven't ordered in the last 6 months (churn risk).
- 121. Employees who share the same department (self join).
- 122. Find duplicate customer emails, if any.
- 123. Products priced identically to another product (self join).
- 124. Most recent order for each customer (window function + filter).
- 125. Customers who have ordered products from every single category ("relational division").
- 126. Product pairs frequently purchased together (market basket analysis).
- 127. Top 3 suppliers by total revenue generated through their products.

- 128.** Products that have never been purchased by anyone (NOT EXISTS).
- 129.** Employees who have never handled a cancelled order.
- 130.** Impact of discounts on total revenue (actual revenue vs. revenue without any discount).
- 131.** Day of the week with the highest number of orders.
- 132.** Suppliers with zero discontinued products (all active).
- 133.** Top 5 customers by number of distinct product categories purchased.
- 134.** Combine cancelled and returned orders into a single list (UNION).
- 135.** Products in the top price quartile of their category (window function).
- 136.** Employees ranked within their own department by salary (DENSE_RANK).
- 137.** Rank all employees by total sales revenue handled.
- 138.** Customers with the highest average order value (AOV).
- 139.** Average time between order date and ship date, per employee.
- 140.** Customers who reviewed a product they actually purchased (EXISTS).
- 141.** Create a view summarizing product performance.
- 142.** Products priced higher than their own category's average price (correlated subquery).
- 143.** Products with below-average ratings but above-average sales volume (cross-metric analysis).
- 144.** Create a view summarizing each customer's order activity.
- 145.** Products ordered more than 50 times in total, with their category name.
- 146.** Number of orders placed by customers in each country during 2025.
- 147.** Each employee's total orders handled and their average order value.
- 148.** Comprehensive order summary report (multijoin + CASE + aggregation).
- 149.** RFM analysis (Recency, Frequency, Monetary) per customer.
- 150.** Find the top 2 best-selling products in each category, by total revenue.